

Water Heaters

One appliance, one safety valve, and the numbers the exam asks for more often than any others in this topic.

The numbers here

Water heater — maximum setting, to prevent scalding	120 degrees F
T&P relief valve — discharge temperature	210 degrees F
T&P relief valve — discharge pressure	150 psi
TPR discharge pipe size	3/4 in
TPR discharge tube — terminates above the floor	6 in
Electric water heater circuit	240 V / 30 amp

Keep the two temperatures apart. 120 degrees F is where the thermostat is set so nobody is scalded at the tap. 210 degrees F is where the safety valve opens, just under boiling, because water turning to steam inside a sealed tank is what makes a water heater explode.

YOU HAVE MISSED THIS 2 TIMES

The safety device that should be attached to the pressure relief valve on a water heater is:

An extension pipe.

The valve on its own would discharge scalding water at head height. The extension pipe carries it down to six inches above the floor, where it can escape without burning whoever is standing there — and where a leak is still visible. The pipe is never threaded at its end and never valved.

Inside the tank

Dip tube	The component introducing cold water near the bottom of the tank, so incoming cold does not mix straight into the hot water leaving from the top.
Lower element failed	The most likely sign is some hot water at first, then warm water at all locations. The upper element still heats the top of the tank, so you get a short burst of genuinely hot water and then lukewarm.
Barometric damper	What a gas water heater will NOT have — it belongs to oil-fired equipment. A gas water heater uses a draft hood.

Say it back

Cover the page. Answer each one out loud.

1. The two water heater temperatures, and what each one is for.
2. T&P valve discharge pressure?
3. What attaches to the relief valve, where does it end, and why?
4. Discharge pipe size, and the circuit for an electric water heater?
5. What does a dip tube do?
6. Some hot water then warm everywhere — which element has failed?
7. Which draft component does a gas water heater not have?